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Greatly improved piezoelectricity and thermal stability of (Na, Sm) Co-doped $CaBi_2Nb_2O_9$ ceramics

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A B S T R A C T

Calcium bismuth niobate ($CaBi_2Nb_2O_9$) is regarded as one of the most potential high-temperature piezoelectric materials owing to its highest Curie point in bismuth layer-structured ferroelectrics. Nevertheless, low piezoelectric coefficient and low resistivity at high temperature considerably restrict its development as key electronic components. Herein, markedly improved piezoelectric properties and DC resistivity of $CaBi_2Nb_2O_9$ ceramics through Na^+ and Sm^{3+} co-doping are reported. The nominal compositions $Ca_{1-2x}(Na, Sm)_xBi_2Nb_2O_9$ (x = 0, 0.01, 0.025, and 0.05) ceramics have been prepared *via* the conventional solid state method. An optimum composition of $Ca_{0.95}(Na, Sm)_{0.025}Bi_2Nb_2O_9$ is obtained, which possesses a high Curie point of ~949 °C, a piezoelectric coefficient of ~12.8 pC/N, and a DC electrical resistivity at 500 °C of ~4 × 10^7 Ω·cm. The improved d_{33} is probably ascribed to the reduction in domain size and the increase in domain wall density caused by the reduced grain size. More importantly, after annealing at 900 °C for 2 h, the piezoelectric coefficient still maintains about 90% of the initial d_{33} value, which displays a significant improvement compared to pure $CaBi_2Nb_2O_9$ ceramic with only 44% of the initial d_{33} value. This work exhibits a feasible approach to simultaneously obtain high piezoelectric property and thermal stability in $CaBi_2Nb_2O_9$ ceramics by Na^+/Sm^{3+} co-doping.

1. Introduction

Piezoelectric materials are capable of the mutual conversion between mechanical and electrical energy, and are widely used in the fields of transducers, sensors, filters, and actuators [1,2]. However, with the rapid development of science and technology, electronic and electrical equipment have put forward higher requirements on the performance and operating conditions of piezoelectric devices. Under some special high-temperature circumstances (especially above 500 °C), there is an urgent need for piezoelectric devices, such as vibration sensors used for the detection of the working status of aero engines, piezoelectric injectors used for advanced automotive electronic fuel injection system and high-power ultrasonic transducers [3–5]. The most widely studied $PbZrTiO_3$-based [6,7]、$K_{0.5}Na_{0.5}NbO_3$-based [8,9]、and $BaTiO_3$-based [10–12] piezoelectric materials with perovskite structure possess large piezoelectric coefficient, but their low T_C of <500 °C limits their applications at high temperature. $BiFeO_3$–$BaTiO_3$-based piezoelectric materials with Curie temperatures beyond 500 °C also have good piezoelectric properties which seem to be expected to be useable at 500 °C [13,14]. However, the tendency of Fe to change its valence state at high temperature causes it to have high leakage current at high temperature [15,16]. This system will not be considered for use at high temperature until this problem is solved. Piezoelectric materials with higher Curie temperature and better thermal stability of piezoelectric properties are eager to meet the demand of high temperature applications.

Bismuth layer-structured ferroelectrics (BLSFs) are also known as Aurivillius phase compounds, most of which exhibit some advantages of high T_C, low aging rate, and high resistivity, *etc* [17,18]. The structure can be expressed by a chemical formula $(Bi_2O_2)^{2+}(A_{m-1}B_mO_{3m+1})^{2-}$, where A is a series of mono-, di-, trivalent ions or their mixture that occupies 12-coordination sites; B represents octameric coordination

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transition metal cations; m represents the number of octahedral layers in the perovskite slab [19,20]. Compared to other BLSFs such as $Bi_4Ti_3O_{12}$ ($T_C \sim 657$ °C, $d_{33} \sim 7.3$ pC/N) [21–23], $CaBi_4Ti_4O_{15}$ ($T_C \sim 790$ °C, $d_{33} \sim 9$ pC/N) [24,25], $Na_{0.5}Bi_{2.5}Nb_2O_9$ ($T_C \sim 770$ °C, $d_{33} \sim 16$ pC/N) [26–28], $CaBi_2Nb_2O_9$ (CBN) is the most promising due to its highest Curie temperature ($T_C \sim 949$ °C) in BLSFs [29], which indicates that it is able to operate at higher temperature. However, the spontaneous polarization of $CaBi_2Nb_2O_9$ is restricted in the *a-b* plane and the high coercive field make it difficult to obtain excellent piezoelectric properties by artificial poling treatment, which is a common problem for BLSFs [18,30]. Grain orientation technique and chemical doping are two major methods to enhance the piezoelectric property of $CaBi_2Nb_2O_9$ ceramics. Grain orientation techniques, including template grain growth method (TGG), spark plasma sintering (SPS), enable in a significant increase in d_{33} of $CaBi_2Nb_2O_9$ ceramics [29,31,32]. Whereas, these methods are complex, low yield, low qualification rate of finished products, and more expensive than the solid state method [33]. Chemical doping method has these advantages of simple process, high yield, and low cost [29], which is more appropriate for large-scale production and commercial applications.

In order to improve the piezoelectric properties of $CaBi_2Nb_2O_9$ ceramics, several doping strategies were implemented in previous studies. High-valent ions such as Ce^{3+} [34] and La^{3+} [35] at A sites as the donor dopant were proved to be effective to depress the generation of oxygen vacancies. The reduced oxygen vacancy concentration leads to increased resistivity and lower coercivity field, which is conducive to the improvement of piezoelectric properties. However, the effect of this doping strategy seems to be limited from the published works (generally d_{33} does not exceed 15 pC/N), such as 10.3 pC/N for Ce^{3+} doped CBN ceramics [34], 13 pC/N for La^{3+} doped CBN ceramics [35].

The donor-acceptor co-doping strategy was reported to be a simple and effective way to significantly improve the piezoelectric coefficient and its thermal stability, which is powerfully demonstrated in $BaTiO_3$ (BT) system [10–12]. In BT system, not only the improvement of piezoelectric properties but also its thermal stability has been achieved by Li^+/Al^{3+} [11,12], Li^+/La^{3+} [10], and Na^+/Er^{3+} [36] co-doping. It has been revealed that the donor and acceptor ions at A sites in BT tend to form ion pairs at adjacent A sites, which will produce the following two effects: first, the lattice distortion caused by the difference in ion radii between doped ions and original ions is beneficial to the improvement of piezoelectric properties; second, defect dipoles caused by the ion pairs are also favorable to the improvement of the piezoelectric properties and its thermal stability [10,11]. Considering the existence of similar perovskite layers in BLSFs, this co-doping strategy has also been employed to improve their piezoelectric properties and its thermal stability. For example, Li^+/Ce^{3+} and K^+/Ce^{3+} ion pairs were respectively used in $BaBi_4Ti_4O_{15}$–$Bi_4Ti_3O_{12}$ [37] and $Bi_4Ti_3O_{12}$ [38], resulting in a significant increase of its piezoelectric coefficient and a good thermal stability. And this strategy was proved to be effective in CBN system as well. The introduction of Na^+/Ce^{3+} ion pairs to CBN also led to a higher improvement in piezoelectric coefficient d_{33} (~16 pC/N) than that (~10.3 pC/N) of Ce^{3+} doped CBN [34,39]. It is reported that the improvement of piezoelectric properties usually is contributed by two aspects. First, the introduction of ion pairs leads to a decrease in the orthorhombicity of CBN triggered by the tilting of $[NbO_6]$ octahedron along a or b axis due to the difference in the ionic radii between the doped and original ions, which leads to enhanced spontaneous polarization in CBN [34,39–41]. In addition, fine domain due to ion-pair introduction has also been observed, which is generally considered as having a positive effect on the piezoelectric properties [23,42].

In previous study, Sm^{3+} has been utilized to modify CBN, however, dissatisfactory result (d_{33} is only 11.5 pC/N) was obtained [31]. In this article, Na^+/Sm^{3+} ion pairs, in which Na^+ acts as the acceptor dopant while Sm^{3+} acts as the donor dopant, were exploited to modify CBN to obtain a better piezoelectric property and thermal stability. Better piezoelectric properties ($d_{33} \sim 12.8$ pC/N) and resistivity ($\rho \sim 4 \times 10^7$ Ω·cm at 500 °C) than those of only Sm^{3+}-doped CBN ceramics were obtained in Na^+/Sm^{3+} co-doped CBN ceramics. The effect of Na^+/Sm^{3+} co-doping on microstructure, defects as well as electrical properties was systematically investigated.

2. Results and discussion

The XRD data of CBNNS-x are given in Fig. 1(a). The experiment results prove the diffraction peaks of CBNNS-x ceramics are line with the PDF card of JCPDS#49–0608, indicating that all samples are only a phase with the space group of $A2_1am$ [20]. The diffraction peak related to the (115) plane is the highest peak in all samples, which corroborates the principle that the most intense diffraction peak of Aurivillius phase compounds is the peak corresponding to the ($11\underline{2m+1}$) plane [43].

To analyze the structures of CBNNS-x, XRD Rietveld refinements were performed with GSAS software, as shown in Fig. 1(d–g) and Tables S1–4. The R-factors (R_{wp} and R_p) and goodness (χ^2) of the fit values show the well-fitting between the test data and fitting values. The change of cell parameters (a, b, and c) and a/b ratio *versus* doping contents is demonstrated in Fig. 1(b and c). Fig. 1(b) shows that there is no significant change in a and b with the growth of doping content, while c displays a significant drop. Fig. 1(c) shows that the a/b value decreases with x growing. It reveals that (Na^+, Sm^{3+}) doping reduces the orthorhombicity of $CaBi_2Nb_2O_9$ and its tetragonality increases [44]. The variation in cell parameter and the orthorhombicity of $CaBi_2Nb_2O_9$ with increasing doping content is induced by the tilting of the oxygen octahedra around Ca^{2+} along a and b axis due to the difference in the ionic radii between Ca^{2+} and Na^+/Sm^{3+}. Similar results have been reported, and this change in lattice triggered by the introduction of Na^+/Sm^{3+} is generally regarded to exert a positive effect to piezoelectric properties as an intrinsic contribution [40,41].

The XPS measurement of CBNNS-x ceramics was conducted to estimate the oxygen vacancy level, as shown in Fig. 2(a1-a4). The photoemission of O 1s of CBNNS-x ceramics shows a clear peak splitting phenomenon and is successfully fitted by using XPS peaks software. The peak located at 529.69 eV represents lattice oxygen (O_L) in ceramics, while the peak located at the binding energy of 531.2 eV is the oxygen-deficient region (oxygen vacancy, V_O). In general, the ratio of the V_O area to the O_L area in the sample is used to quantify the oxygen vacancy concentration [45,46]. For CBNNS-x ceramics, Bi and Na elements are lost by volatilizing during high-temperature sintering, leading to A-site ion defects. And oxygen vacancies are generated to maintain electroneutrality. In Fig. 2(a1-a4), the ratio of the oxygen vacancy to the lattice oxygen in CBNNS-x ceramics decreases from 0.3220 to 0.2886 and then increases to 0.3014 as the doping amount x increases. The results indicate that moderate doping may reduce the oxygen vacancy concentration. While x exceeds 0.025, the oxygen vacancy concentration becomes higher instead, which may be caused by the increase of volatile elements Na in the ceramics [37].

SEM images, average grain size in length and relative densities of CBNNS-x ceramics are displayed in Fig. 2(b1-b4) and Fig. 2(c and d). The CBNNS-x ceramics possess plate-like grains, which shows strong anisotropy due to the fact that grains grow faster along the *a-b* plane than along other directions [47]. The highest node density in the *a-b* plane than other planes in BLSFs is responsible for the fastest growth speed along *a-b* plane according to the Law of Bravais [48] and Gibbs-Wulff growth law [49]. As depicted in Fig. 2(c), as x increasing, the average grain size of CBNNS-x ceramics declines from 7.7 μm to 5.4 μm and then it markedly increases to 9.0 μm. This may be affected by the variation of defects in CBNNS-x ceramics. It is generally believed that oxygen vacancy with lower activation energy allows for an increase in substance transport efficiency during sintering, which promotes the grain growth [21]. Therefore, the compositions with lower oxygen vacancy concentration exhibit a smaller grain size. In addition, all CBNNS-x ceramics maintain a high relative density of more than 98%, as shown in Fig. 2(d). In $CaBi_2Nb_2O_9$-based ceramics, smaller grain size usually contributes to the

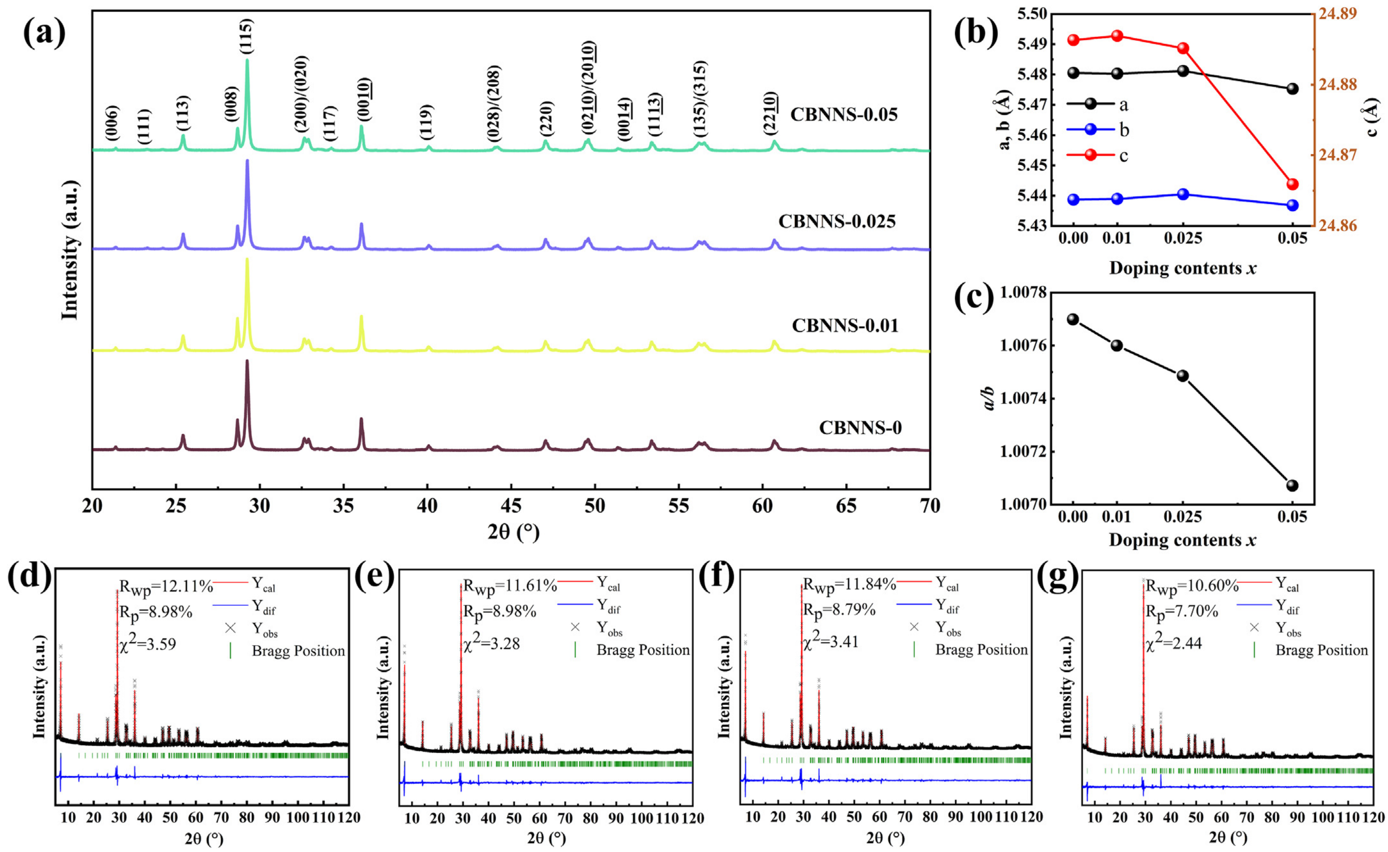

Fig. 1. (a)X-ray diffraction patterns of CBNNS-*x* ceramics. (b) Variation of cell parameters and (c) *a/b* ratio of CBNNS-*x* ceramics *versus* doping contents. Rietveld refinement plots of XRD patterns of CBNNS-0 (d), 0.01 (e), 0.025 (f), 0.05(g) ceramics.

piezoelectric response due to the decrease in domain width and the increase in domain wall density [50]. It is expected to obtain an optimized piezoelectric response in CBNNS-0.025 ceramic.

The TEM image of CBNNS-0.025 ceramic is demonstrated in Fig. 2(e), in which domain structure appearing as alternately light and dark stripes is visible in these regions ②, ③ and ④, which is similar to the observation in $Bi_4Ti_3O_{12}$ ceramics [22,23]. Such a high-density nanodomain structure is favorable to achieve high piezoelectric properties in ferroelectric ceramics [51]. The SAED of the region ① in the left edge of Fig. 2(e) is exhibited in Fig. 2(f), which can be indexed by the [110] zone axis in orthorhombic cell. And similar results were also observed in $Bi_4Ti_3O_{12}$ [52]. In Fig. 2(g), high-resolution TEM image corresponding to the region ① provides the average interplanar spacing of the (−115) and (002) crystal planes, ~0.3047 nm and ~1.2507 nm, respectively. Additionally, the inset in Fig. 2(g) shows a cell of $CaBi_2Nb_2O_9$ along the [100] direction, which is consistent with the result of the HRTEM image.

As a typical characteristic of ferroelectric materials, polarization-field (*P-E*) hysteresis loop can reveal polarization reversal behaviors. Fig. 3(a1-a5) and Figs. S1(a–c) exhibit the *P-E* and *I-E* hysteresis loops of CBNNS-*x* ceramics at room temperature and 200 °C. There exists no switching current peak in the *I-E* hysteresis loops of CBNNS-*x* ceramic at room temperature, which shows that the polarization switching is not easy to occur. When the temperature comes 200 °C, the *P-E* hysteresis loops of all samples become more open and switching current peaks (I_{max}) are appeared at E_{Imax} in *I-E* hysteresis loops. The remnant polarization $2P_r$ represents the absolute value of the difference between the two intersection points of the *P-E* loop and the *E*=0 axis. And the remnant polarization $2P_r$ of CBNNS-*x* ceramics at 200 °C is displayed in Fig. 3(b). It can be observed that both the $2P_r$ and E_{Imax} increases with increasing *x* from 0 to 0.025 and decreases when x reaches 0.05. E_{Imax} usually reflects the resistance to domain switching in ferroelectric materials, and the larger the E_{Imax}, the more difficult the domain switching [53]. It can be found that the compositions with a smaller grain size exhibit larger E_{Imax}, which is probably ascribed to the enhanced restriction on domain switching by more grain boundaries [54,55]. Since the applied electric field is much higher than E_{Imax}, the $2P_r$ does not seem to be affected by higher E_{Imax}. Instead, higher E_{Imax} will prevent the domain switching back after the electric field is removed, favoring the enhancement of the $2P_r$. In addition, the enhancement of the spontaneous polarization caused by the reduced orthogonality will also facilitate the enhancement of the remnant polarization $2P_r$ [44].

In general, the piezoelectric coefficient of poled ferroelectric ceramics can be suggested as: $d_{33}=2\varepsilon_r P_r S$, which is directly proportional to dielectric constant, remnant polarization and strain [56–59]. It should be noted that the contributions to these parameters include two aspects: intrinsic and extrinsic contributions. The intrinsic contributions originate from the field-induced polarization elongation/rotation or deformation of a single domain, while the extrinsic ones mainly arise from the displacement of domain walls and interphase boundaries. According to the above-mentioned equation, it can be concluded that the increase in ε_r (poled samples) and P_r is beneficial for the improvement of d_{33}. In order to investigate the contributions to the piezoelectric performance of CBNNS-*x* ceramics, the frequency-dependent ε_r of poled CBNNS-*x* and d_{33} in room temperature are measured, as displayed in Fig. 3(c1-c4) and Fig. 3(d), respectively. For every composition, the ε_r of ceramics gradually decreases with increasing poling electric field because poling process leads to irreversible domain wall switching and results in a decrease in the density of the non-180° ferroelectric domain walls, which reduces the dielectric response [60]. According to Fig. 3(d and e), it can be found that compositions with higher piezoelectric properties have a lower relative permittivity ε_r but a higher remnant polarization $2P_r$, which indicates that the enhancement of d_{33} in CBN ceramics is mainly contributed by the

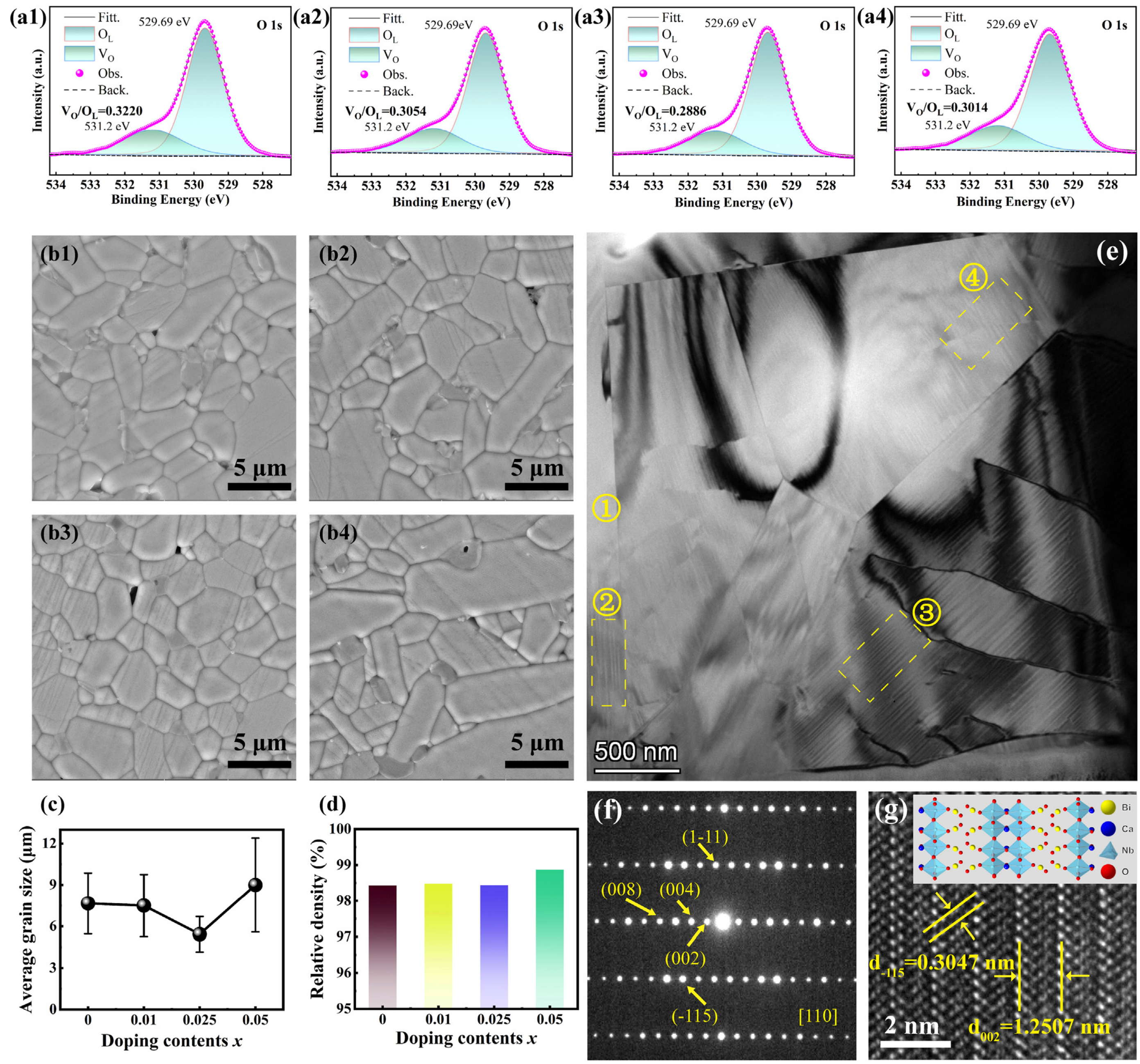

Fig. 2. XPS O 1s spectra of CBNNS-0 (a1), CBNNS-0.01 (a2), CBNNS-0.025 (a3), CBNNS-0.05 (a4) ceramics. The SEM images (b1-b4) of CBNNS-*x* ceramics after polishing and thermal etching, average grain size in length (c) and relative density (d) varying with doping contents *x*. TEM images of CBNNS-0.025 ceramics: (e) overview bright-field image, (f) selected area electron diffraction (SAED) pattern and (g) HRTEM image taken from the region ① in (e).

significant increase in $2P_r$. This may be because the slight change (less than 6%) in relative permittivity with doping content has a weaker effect on piezoelectric coefficient compared to the significant change (close to 30%) in remnant polarization $2P_r$.

The variation of dielectric constant ε_r and dielectric loss tanδ with temperature for the CBNNS-*x* ceramics, as well as the Curie point T_C and tanδ at 500 °C varying with doping amount are exhibited respectively in Fig. 4(a–c). For BLSFs with an even layer, the ferroelectric phase corresponds to orthorhombic phase, while paraelectric phase to tetragonal phase [61]. The T_C of $CaBi_2Nb_2O_9$ ceramics is the temperature at which the orthogonal phase transforms into the tetragonal phase. As depicted in Fig. 4(b), the Curie point exhibits a slight decrease as *x* increases. The reduction in T_C is relevant to the decrease of orthorhombicity caused by the introduction of Na^+ and Sm^{3+}. The lower the orthorhombicity, the lower the T_C [53,62]. The variation of T_C values is coincided with that of a/b values shown in Fig. 1(c). From Fig. 4(c), the tanδ of all CBNNS-*x* ceramics is lower than 0.015 at 500 °C. The variation of tanδ may be closely involved with oxygen vacancy concentration, and the tanδ of CBNNS-0.025 ceramics is markedly reduced compared to the pure sample, which indicates that the CBNNS-0.025 ceramics have a lower loss and is beneficial for its application in extreme environments.

To further investigate the resistivity of CBNNS-*x* ceramics under a high-temperature environment, the temperature-dependent DC resistivity ρ of CBNNS-*x* ceramics is shown in Fig. 4(d). The DC resistivity of all samples decreases with increasing temperature, suggesting that the conducting charge carriers of CBNNS-*x* ceramics are thermally activated carriers. The DC resistivity of the CBNNS-*x* ceramics at 500 and 700 °C is demonstrated in Fig. 4(e). The resistivity of all samples was maintained

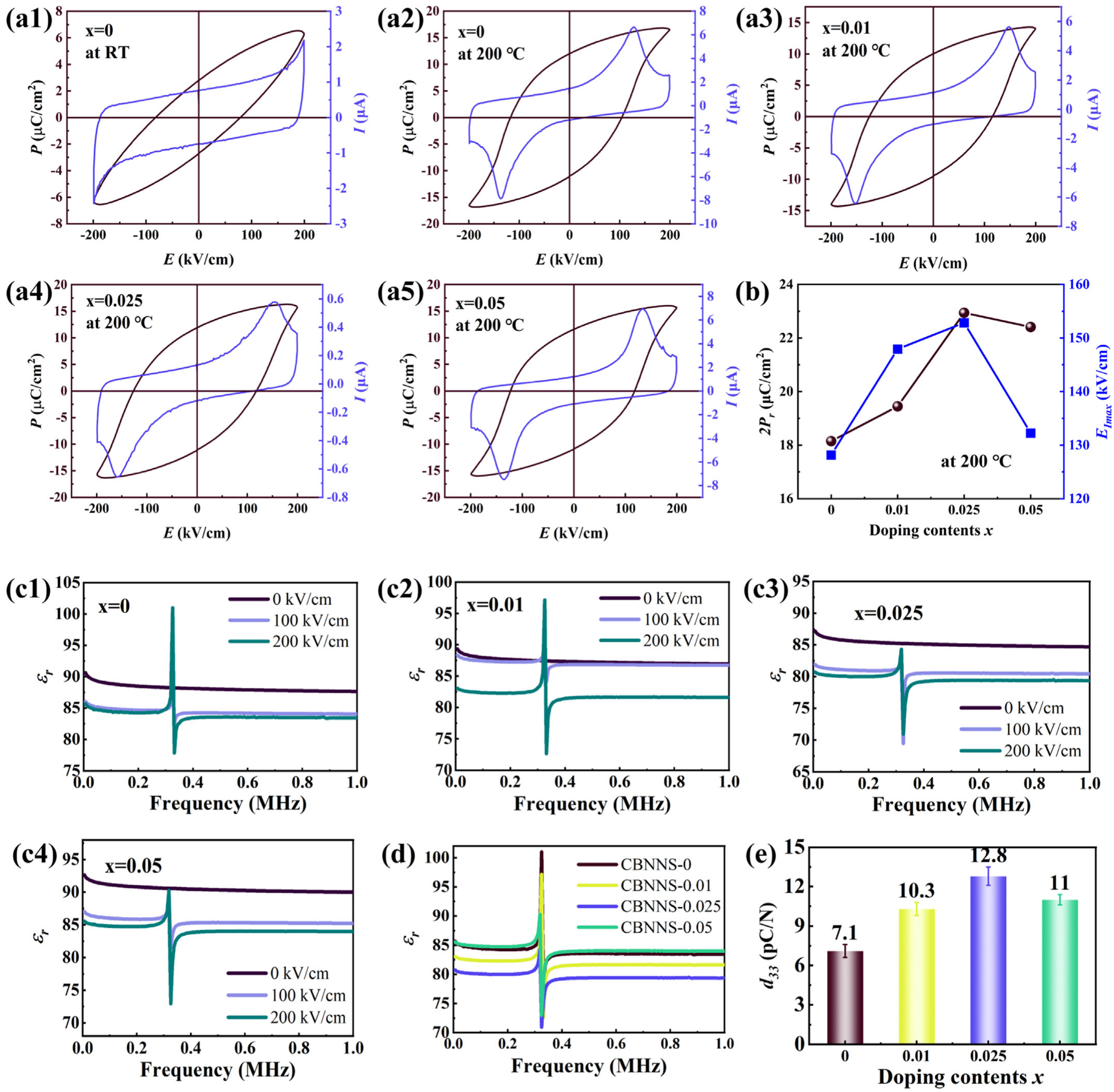

Fig. 3. *P-E* and *I-E* loops of CBNNS-0 ceramic measured at room temperature (a1) and CBNNS-*x* ceramics measured at 200 °C (a2-a5); (f) Variation of $2P_r$ and E_{Imax} as a function of doping contents *x* in CBNNS-*x* ceramics at 200 °C. (c1-c4) Frequency-dependent ε_r of CBNNS-*x* ceramics poled at different electric field. (d) Frequency-dependent ε_r of the CBNNS-*x* ceramics poled at 200 kV/cm, (e) Room-temperature (RT) d_{33} of CBNNS-*x* ceramics.

above 10^7 Ω·cm, and the doped samples all showed a significant increase in resistivity compared to the undoped samples, which facilitates the CBNNS-*x* ceramics to be applied at high temperatures. The increased resistivity is probably connected to the reduced oxygen vacancies and the resulting grain size reduction which induces an increase of grain boundaries with higher resistance [63,64].

The DC resistivity of CBNNS-*x* samples varying with temperature can be expressed by the Arrhenius formula ρ=A exp ($-E_a/kT$), where E_a is the activation energy of per charge carrier migration [65,66]. To find out the conductivity mechanism of samples, the carrier activation energies in the low temperature (200–400 °C) and the high temperature (450–800 °C) obtained by fitting the Arrhenius equation are shown in Fig. 4(f). Carrier activation energy is the energy required for a carrier to escape from original location, so a high activation energy signifies a high resistivity and a low leakage current [67]. The low-temperature activation energy of all samples is smaller than the high-temperature activation energy, which indicates different conductivity mechanisms in different temperature ranges. The activation energies at low temperatures range from 0.9 to 1.02 eV, which are in line with the activation energy of movement of electrons associated with oxygen vacancy, implying that conductivity in low-temperature range is mainly dominated by defects and impurities. At *x*=0.025, the carrier activation energy E_a obtains a maximum value, probably triggered by the drop of oxygen vacancy. The activation energy at high temperatures varies from 1.8 to 2.02 eV, which is about half of the

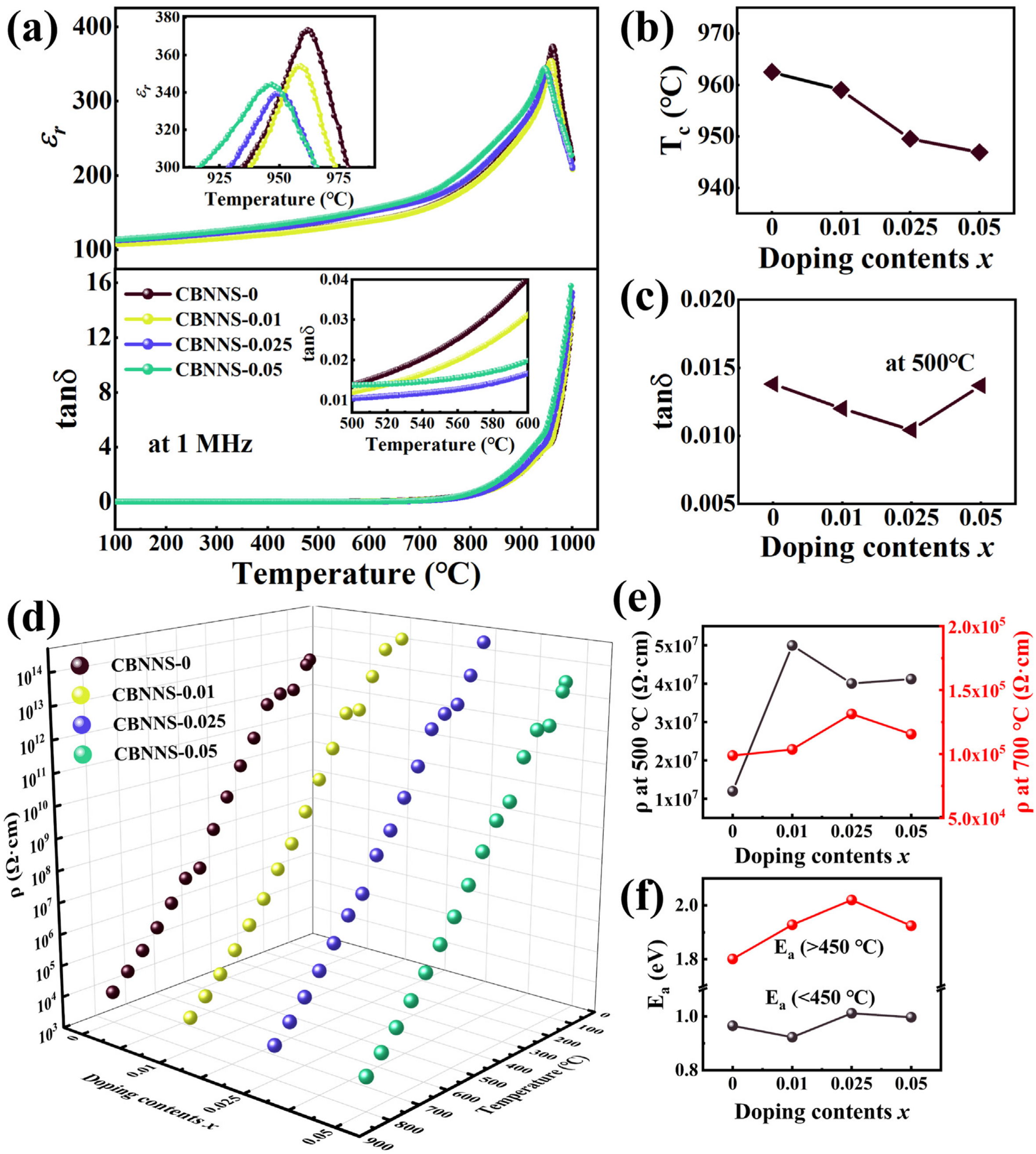

Fig. 4. (a) The variation of dielectric constant and dielectric loss with temperature for CBNNS-*x* ceramics at 1 MHz, (b) the Curie point and (c) tanδ at 500 °C versus doping content *x*. (d) DC resistivity of CBNNS-*x* ceramics as a function of temperature from RT to 800 °C. (e) DC resistivity of CBNNS-*x* ceramics at 500 and 700 °C. (f) Activation energy (E_a) of CBNNS-*x* ceramics at low and high temperature.

band-gap energy ($E_g \sim 3.40$ eV) of $CaBi_2Nb_2O_9$ ceramic and corresponds to the intrinsic carrier conduction mechanism [68]. The intrinsic conduction is associated with the polycrystalline nature of samples [26,69]. The increase in resistivity of doped $CaBi_2Nb_2O_9$ ceramics at 500 °C may be attributed to the higher activation energy in the high temperature region.

In order to test the thermal stability of CBNNS-x ceramics, thermal depolarization was implemented as shown in Fig. 5(a), which determines the maximum service temperature [70–73]. At 400 °C, except for the pure $CaBi_2Nb_2O_9$ ceramics showing a large drop, the d_{33} of doped $CaBi_2Nb_2O_9$ ceramics showed only a slight change. When annealing temperature reaches 900 °C, the pure $CaBi_2Nb_2O_9$ ceramics exhibits a large drop in d_{33}, while all the doped $CaBi_2Nb_2O_9$ ceramics maintain a good temperature stability, and the piezoelectric coefficient d_{33} remain close to 90% of that at room temperature. This also indicates the stability of the d_{33} of $CaBi_2Nb_2O_9$ ceramics is improved by the co-doping of Na^+ and Sm^{3+}. The piezoelectric properties of ceramic materials come from two sources: intrinsic and extrinsic contribution. The intrinsic contribution is caused by the displacement of atoms in the cell, while the extrinsic contribution is induced by the movement of domain walls [58]. Defect dipoles consisting of oxygen vacancies and cation vacancies are able to orientate with domains switching under the electric field during poling treatment, which has a pinning effect on the switched domains and makes them more stable [10,11]. However, at high temperatures, the oxygen vacancies with low activation energy tend to migrate, which makes defect dipoles thermally degrade and lose their stabilizing effect on domains, thus affecting the thermal stability of the piezoelectric properties [74]. The thermal degradation of defect dipoles at high

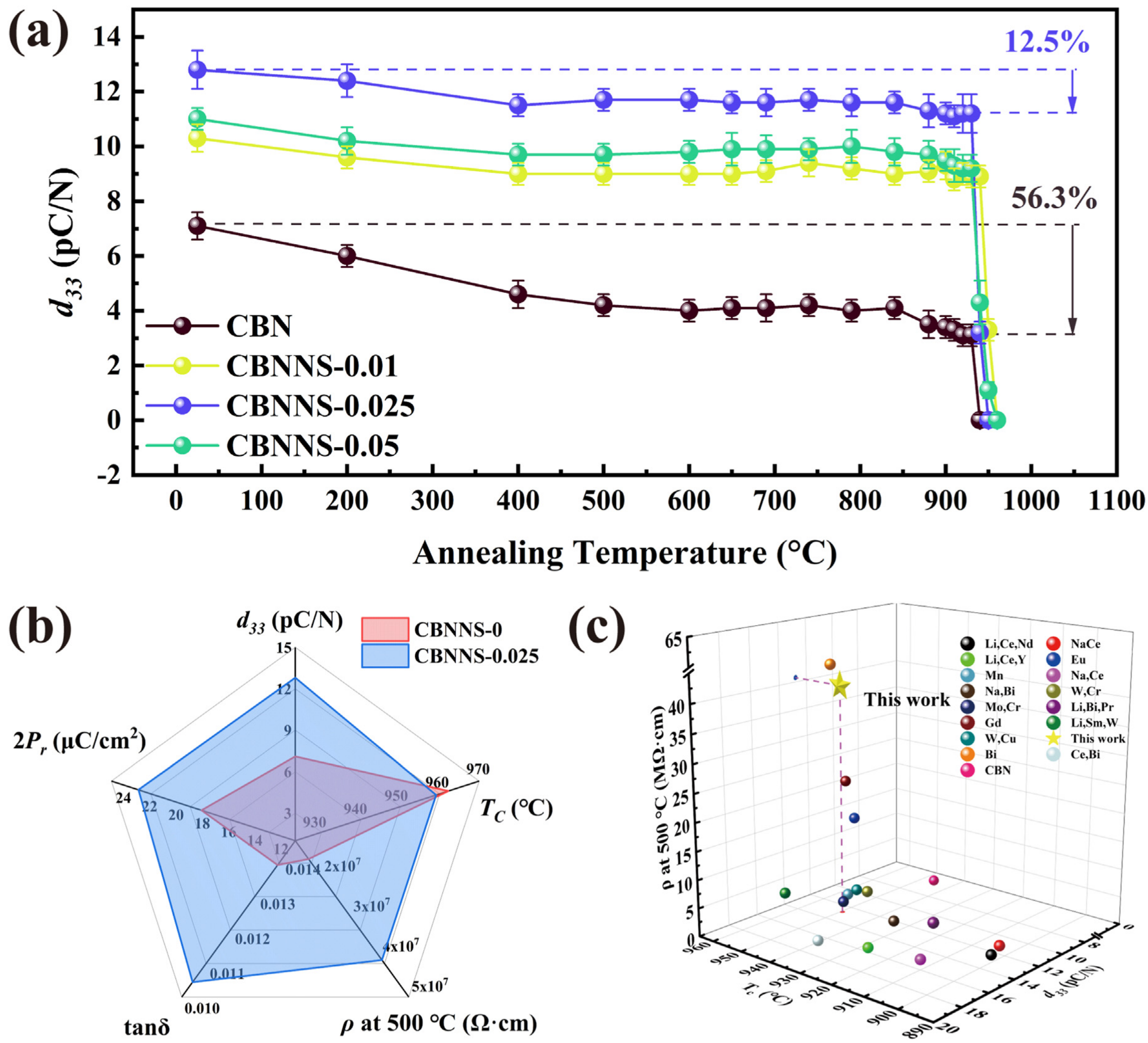

Fig. 5. (a) Thermal depolarization behavior of CBNNS-x ceramics. (b) Comparison of comprehensive properties of the CBNNS-0 and CBNNS-0.025 ceramic samples. (c) Correlations of T_c, d_{33}, and ρ at 500 °C of $CaBi_2Nb_2O_9$-based ceramics with different doping elements.

temperatures has the weakest influence on CBNNS-0.025 ceramic with the lowest oxygen vacancy concentration, which results in higher thermal stability of CBNNS-0.025 ceramic, and the strongest influence on CBNNS-0 ceramic with the highest oxygen vacancy concentration, making it the worst thermal stability. The comparison of comprehensive properties of the CBNNS-0 and CBNNS-0.025 ceramic samples and the correlations of T_C, d_{33}, and ρ at 500 °C of $CaBi_2Nb_2O_9$-based ceramics with different doping elements are shown in Fig. 5(b and c). By co-doping with Na^+ and Sm^{3+}, the comprehensive properties of $CaBi_2Nb_2O_9$ are fully enhanced and higher resistivity is obtained compared to other elemental doping, which is favorable for applications at high temperatures.

3. Conclusion

CBNNS-x ceramics were prepared by the conventional solid-state method, and the phase structure, microscopic morphology, and electrical properties were systematically characterized. All samples were single phase and their grain morphology showed a typical flat-plate structure. And striped domains and layered structure are clearly observed respectively in overview bright field image and HRTEM image. The Na^+/Sm^{3+} doping reduced the orthorhombicity of $CaBi_2Nb_2O_9$ ceramics, and the piezoelectric properties of the doped $CaBi_2Nb_2O_9$ ceramics are significantly enhanced, which was attributed to the further separation of the positive and negative charge centers due to the increased degree of distortion and the high resistivity allowing them to be poled under higher electric fields. In addition, thermal depolarization performance tests showed that the piezoelectric coefficient d_{33} of CBNNS-0.025 ceramics remained close to 90% after annealing at 900 °C. The good thermal stability of $CaBi_2Nb_2O_9$ ceramics in terms of piezoelectric properties makes them highly competitive in high-temperature environments above 500 °C.

4. Experimental

The compositions of $Ca_{1-x}(Na, Sm)_xBi_2Nb_2O_9$ abbreviated as CBNNS-x (x=0, 0.01, 0.025, and 0.05)) ceramics were fabricated *via* the conventional solid state method. $CaCO_3$ (99.9%), Na_2CO_3 (99.9%), Sm_2O_3 (99.99%), Bi_2O_3 (99.9%) and Nb_2O_5 (99.9%) were utilized as raw materials to synthesize the above-mentioned compounds. The raw materials were weighed in accordance with the stoichiometric ratio, and then mixed by ball milling for 8 h using anhydrous ethanol as solvent and zirconia balls with different size ratios as the milling medium. After being dried, these mixtures were calcined at 925 °C for 2 h. After this, powders after being calcined were re-milled for 24 h in the same condition, which were mixed using 5 wt % polyvinyl alcohol (PVA) aqueous solution as a binder after drying and pressed into pellets. Disk-shaped samples were heated to 550 °C for 2 h to burn out the PVA binder and sintered at 1150 °C for 2 h. After being sintered and polished, specimens were coated with platinum/silver electrodes for electrical measurements.

The phase structure of CBNNS-*x* ceramics was examined by an X-ray diffraction meter (XRD, Advance D8, Bruker, Switzerland) using Cu K_α radiation (λ=0.15406 nm). The grain morphology of CBNNS-*x* ceramics was observed by field-emission scanning electron microscope (SEM, Nova Nano SEM230, FEI Electron Optics B.V, Czech Republic). The densities of CBNNS-*x* ceramics were measured using the Archimedes method. The *P-E* and *I-E* hysteresis loops were measured using ferroelectric tester (aixACCT Analyzer TF3000). A transmission electron microscope (Titan G2 60–300, FEI, USA) was used to obtain transmission electron microscopy (TEM) and high-resolution transmission electron microscopy (HRTEM) images, as well as selected area electron diffraction (SAED) patterns. For TEM characterization, samples were processed by focused ion beam (FIB) milling (Helios Nanolab G3 UC, FEI, USA). Chemical composition and the valence states of the elements were analyzed *via* X-ray photoelectron spectroscopy (XPS, ESCALAB250Xi, Thermo Fisher Scientific, USA). Temperature-dependent permittivity and dielectric loss were measured by using a dielectric measurement system (TZDM, RT-1000) connected to a programmed furnace with a step of 5 °C/min. The room temperature frequency-dependent dielectric performance of CBNNS-*x* ceramics poled in different poling electric fields was measured using an LCR meter (Agilent 4294A). Variable-temperature direct-current (DC) resistance of CBNNS-*x* ceramics was measured using a resistivity tester with a test voltage of 100 V (RMS-1000P, Partulab, China). Sintered ceramics were poled in silicone oil at 200 °C under different DC electric field (200–250 kV/cm) for about 30–60 min. The piezoelectric coefficient d_{33} of samples which were poled by direct current field was measured at room temperature by using a quasi-static piezometer (ZJ-6, Institute of Acoustics, Chinese Academic of Science, Beijing, China). For evaluating thermal stability, thermal depolarization experiments were implemented by heating the poled samples from room temperature to selected temperatures and holding them at the selected temperature for 2 h and then naturally cooling to room temperature to measure d_{33}. This procedure was repeated many times from room temperature to Curie point.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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References

[1] G. Liu, S. Ren, C. Wu, D. Wang, F. Li, J. Wu, Q. Chen, Enhanced thermal stability of (NaCe)-multidoped $CaBi_2Nb_2O_9$ by A-site vacancies-induced pseudo-tetragonal distortion, J. Am. Ceram. Soc. 101 (2018) 4615–4626, https://doi.org/10.1111/jace.15724.

[2] W. Chen, R. Zhang, Y. Sun, J. Wang, Y. Fan, B. Liu, Preparation, properties, and electronic applications of 2D Bi_2O_2Se, Adv. Powder Mater. 2 (2023), 100080, https://doi.org/10.1016/j.apmate.2022.100080.

[3] U. De, K.R. Sahu, A. De, Ferroelectric materials for high temperature piezoelectric applications, Solid State Phenom. 232 (2015) 235–278. https://doi.org/10.4028/www.scientific.net/SSP.232.235.

[4] S. Zhang, F. Yu, Piezoelectric materials for high temperature sensors, J. Am. Ceram. Soc. 94 (2011) 3153–3170, https://doi.org/10.1111/j.1551-2916.2011.04792.x.

[5] R.C. Turner, P.A. Fuierer, R.E. Newnham, T.R. Shrout, Materials for high temperature acoustic and vibration sensors: A review, Appl. Acoust. 41 (1994) 299–324, https://doi.org/10.1016/0003-682X(94)90091-4.

[6] K. Okazaki, K. Nagata, Effects of grain size and porosity on electrical and optical properties of PLZT ceramics, J. Am. Ceram. Soc. 56 (1973) 82–86, https://doi.org/10.1111/j.1151-2916.1973.tb12363.x.

[7] T.B. Weston, A.H. Webster, V.M. McNamara, Lead zirconate-lead titanate piezoelectric ceramics with iron oxide additions, J. Am. Ceram. Soc. 52 (1969) 253–257, https://doi.org/10.1111/j.1151-2916.1969.tb09178.x.

[8] T. Zheng, J. Wu, D. Xiao, J. Zhu, Recent development in lead-free perovskite piezoelectric bulk materials, Prog. Mater. Sci. 98 (2018) 552–624, https://doi.org/10.1016/j.pmatsci.2018.06.002.

[9] P. Li, J. Zhai, B. Shen, S. Zhang, X. Li, F. Zhu, X. Zhang, Ultrahigh piezoelectric properties in textured $(K,Na)NbO_3$-based lead-free ceramics, Adv. Mater. 30 (2018), 1705171, https://doi.org/10.1002/adma.201705171.

[10] R. Song, Y. Zhao, W. Li, Y. Yu, J. Sheng, Z. Li, Y. Zhang, H. Xia, W.D. Fei, High temperature stability and mechanical quality factor of donor-acceptor co-doped $BaTiO_3$ piezoelectrics, Acta Mater. 181 (2019) 200–206, https://doi.org/10.1016/j.actamat.2019.09.045.

[11] Y. Feng, W.L. Li, D. Xu, Y.L. Qiao, Y. Yu, Y. Zhao, W.D. Fei, Defect engineering of lead-free piezoelectrics with high piezoelectric properties and temperature-stability, ACS Appl. Mater. Interfaces 8 (2016) 9231–9241, https://doi.org/10.1021/acsami.6b01539.

[12] D. Xu, W.L. Li, L.D. Wang, W. Wang, W.P. Cao, W.D. Fei, Large piezoelectric properties induced by doping ionic pairs in $BaTiO_3$ ceramics, Acta Mater. 79 (2014) 84–92, https://doi.org/10.1016/j.actamat.2014.07.023.

[13] B. Xun, A. Song, J. Yu, Y. Yin, J.F. Li, B.P. Zhang, Lead-free $BiFeO_3$-$BaTiO_3$ ceramics with high curie temperature: Fine compositional tuning across the phase boundary for high piezoelectric charge and strain coefficients, ACS Appl. Mater. Interfaces 13 (2021) 4192–4202, https://doi.org/10.1021/acsami.0c20381.

[14] Y. Sun, H. Yang, S. Guan, Y. Cao, M. Jiang, X. Liu, Q. Chen, M. Li, J. Xu, Strong piezoelectricity of Li_2CO_3-doped $BiFeO_3$-$BaTiO_3$-$Bi(Zn_{0.5}Ti_{0.5})O_3$ lead-free piezoelectric ceramics with high curie temperature and high temperature stability, J. Alloys Compd. 819 (2020), 153058, https://doi.org/10.1016/j.jallcom.2019.153058.

[15] M.H. Lee, D.J. Kim, J.S. Park, S.W. Kim, T.K. Song, M.-H. Kim, W.J. Kim, D. Do, I.K. Jeong, High-performance lead-free piezoceramics with high curie temperatures, Adv. Mater. 27 (2015) 6976–6982, https://doi.org/10.1002/adma.201502424.

[16] T. Rojac, M. Kosec, B. Budic, N. Setter, D. Damjanovic, Strong ferroelectric domain-wall pinning in $BiFeO_3$ ceramics, J. Appl. Phys. 108 (2010), 074107, https://doi.org/10.1063/1.3490249.

[17] T. Takenaka, K. Sakata, Grain orientation effects on electrical properties of bismuth layer-structured ferroelectric $Pb_{(1-x)}(NaCe)_{x/2}Bi_4Ti_4O_{15}$ solid solution, J. Appl. Phys. 55 (1984) 1092–1099, https://doi.org/10.1063/1.333198.

[18] B. Frit, J.P. Mercurio, The crystal chemistry and dielectric properties of the Aurivillius family of complex bismuth oxides with perovskite-like layered structures, J. Alloys Compd. 188 (1992) 27–35, https://doi.org/10.1016/0925-8388(92)90639-Q.

[19] E.C. Subbarao, A family of ferroelectric bismuth compounds, J. Phys. Chem. Solid. 23 (1962) 665–676, https://doi.org/10.1016/0022-3697(62)90526-7.

[20] R.E. Newnham, R.W. Wolfe, J.F. Dorrian, Structural basis of ferroelectricity in the bismuth titanate family, Mater. Res. Bull. 6 (1971) 1029–1039, https://doi.org/10.1016/0025-5408(71)90082-1.

[21] F. Zhang, Y. Xu, H. Yang, S. Guan, W. Shi, Y. Chen, C. Huang, J. Xing, H. Liu, Q. Chen, Effects of cerium on structures and electrical properties of (Nb, Ta) modified $Bi_4Ti_3O_{12}$ piezoelectric ceramics, J. Am. Ceram. Soc. 105 (2022) 4161–4170, https://doi.org/10.1111/jace.18393.

[22] X. Xie, Z. Zhou, R. Liang, X. Dong, Superior piezoelectricity in bismuth titanate-based lead-free high-temperature piezoceramics via domain engineering, Adv. Electron. Mater. (2022), 2101266, https://doi.org/10.1002/aelm.202101266 n/a.

[23] X. Xie, Z. Zhou, B. Gao, Z. Zhou, R. Liang, X. Dong, Ion-pair engineering-induced high piezoelectricity in $Bi_4Ti_3O_{12}$-based high-temperature piezoceramics, ACS Appl. Mater. Interfaces 14 (2022) 14321–14330, https://doi.org/10.1021/acsami.1c19445.

[24] X. Xie, Z. Zhou, T. Wang, R. Liang, X. Dong, High temperature impedance properties and conduction mechanism of W^{6+}-doped $CaBi_4Ti_4O_{15}$ aurivillius piezoceramics, J. Appl. Phys. 124 (2018), 204101, https://doi.org/10.1063/1.5053741.

[25] H. Yan, C. Li, J. Zhou, W. Zhu, L. He, Y. Song, Y. Yu, Effects of A-site (NaCe) substitution with Na-deficiency on structures and properties of $CaBi_4Ti_4O_{15}$-based high-curie-temperature ceramics, Jpn. J. Appl. Phys. 40 (2001) 6501.

[26] Y. Wu, J. Chen, J. Yuan, J. Xing, Z. Tan, L. Jiang, Q. Chen, J. Zhu, Structure refinements and the influences of A-site vacancies on properties of $Na_{0.5}Bi_{2.5}Nb_2O_9$-based high temperature piezoceramics, J. Appl. Phys. 120 (2016), 194103, https://doi.org/10.1063/1.4967991.

[27] Z. Zhou, Y. Li, S. Hui, X. Dong, Effect of tungsten doping in bismuth-layered $Na_{0.5}Bi_{2.5}Nb_2O_9$ high temperature piezoceramics, Appl. Phys. Lett. 104 (2014), 012904, https://doi.org/10.1063/1.4861460.

[28] C. Long, H. Fan, M. Li, High temperature aurivillius piezoelectrics: The effect of (Li, Ln) modification on the structure and properties of (Li, Ln)$_{0.06}$(Na, Bi)$_{0.44}$$Bi_2Nb_2O_9$ (Ln = Ce, Nd, La and Y), Dalton Trans. 42 (2013) 3561, https://doi.org/10.1039/C2DT32564K.

[29] H. Yan, H. Zhang, R. Ubic, M.J. Reece, J. Liu, Z. Shen, Z. Zhang, A lead-free high-Curie-point ferroelectric ceramic, $CaBi_2Nb_2O_9$, Adv. Mater. 17 (2005) 1261–1265, https://doi.org/10.1002/adma.200401860.

[30] Z. Peng, Q. Chen, D. Liu, Y. Wang, D. Xiao, J. Zhu, Evolution of microstructure and dielectric properties of (LiCe)-doped $Na_{0.5}Bi_{2.5}Nb_2O_9$ Aurivillius type ceramics, Curr. Appl. Phys. 13 (2013) 1183–1187, https://doi.org/10.1016/j.cap.2013.03.008.

[31] H. Chen, X. Guo, Z. Cui, J. Zhai, Effects of A-site sm substitution and textured structure on electric properties of $CaBi_2Nb_2O_9$-based high-curie-temperature ceramics, Int. J. Appl. Ceram. Technol. 10 (2013) E151–E158, https://doi.org/10.1111/ijac.12082.

[32] H.B. Chen, J.W. Zhai, Enhanced piezoelectric properties of $CaBi_2Nb_2O_9$ with eu modification and templated grain growth, Key Eng. Mater. 512–515 (2012) 1367–1371. https://doi.org/10.4028/www.scientific.net/KEM.512-515.1367.

[33] J.G. Wu, X.Y. Gao, J.G. Chen, C.M. Wang, S.J. Zhang, S.X. Dong, Review of high temperature piezoelectric materials, devices, and applications, Acta Phys. Sin. 67 (2018) 207701, https://doi.org/10.7498/aps.67.20181091.

[34] C. Qin, Z.Y. Shen, W.Q. Luo, F. Song, Z. Wang, Y. Li, Microstructure related properties enhancing in Ce-doped $CaBi_2Nb_2O_9$ high temperature piezoelectric ceramics, Mater. Res. Express 6 (2019), 106308, https://doi.org/10.1088/2053-1591/ab3ae6.

[35] X. Zhang, H. Yan, M.J. Reece, Effect of a site substitution on the properties of $CaBi_2Nb_2O_9$ ferroelectric ceramics, J. Am. Ceram. Soc. 91 (2008) 2928–2932, https://doi.org/10.1111/j.1551-2916.2008.02564.x.

[36] J. Li, X. Chai, D. Peng, H. Zou, X. Wang, X. Yao, Largely enhanced electromechanical properties of $BaTiO_3$-($Na_{0.5}Er_{0.5}$)TiO_3 lead-free piezoelectric ceramics, Appl. Phys. Lett. 105 (2014), 082901, https://doi.org/10.1063/1.4893610.

[37] W. Fan, X. Jiang, C. Chen, X. Huang, X. Nie, X. Wang, Electrical properties and curie temperature of Li/Ce co-doped $BaBi_4Ti_4O_{15}$–$Bi_4Ti_3O_{12}$ intergrowth ceramics, Ceram. Int. 48 (2022) 2833–2842, https://doi.org/10.1016/j.ceramint.2021.10.073.

[38] Z.Y. Shen, Z. Zhang, C. Qin, W. Luo, F. Song, Z. Wang, Y. Li, Effect of ($K_{0.5}Ce_{0.5}$) complex on the electrical properties of lead-free $Bi_4Ti_{2.86}W_{0.14}O_{12}$ high-temperature piezoelectric ceramics, J. Mater. Res. 36 (2021) 1134–1141, https://doi.org/10.1557/s43578-020-00078-1.

[39] C.M. Wang, S. Zhang, J.F. Wang, M.L. Zhao, C.L. Wang, Electromechanical properties of calcium bismuth niobate ($CaBi_2Nb_2O_9$) ceramics at elevated temperature, Mater. Chem. Phys. 118 (2009) 21–24, https://doi.org/10.1016/j.matchemphys.2009.06.022.

[40] X. Zeng, F. Cao, Z. Peng, X. Xing, Crystal structure and electrical properties of (Li, Ce, Nd)-multidoped $CaBi_2Nb_2O_9$ high temperature ceramics, Ceram. Int. 44 (2018) 3069–3076, https://doi.org/10.1016/j.ceramint.2017.11.069.

[41] G. Liu, C. Wu, Y. Chen, D. Liang, B. Wang, J. Wu, Q. Chen, Enhanced ferroelectricity of $CaBi_2Nb_2O_9$-based high-temperature piezoceramics by pseudo-tetragonal distortion, Ceram. Int. 44 (2018) 5880–5885, https://doi.org/10.1016/j.ceramint.2017.12.019.

[42] F. Wang, X. Li, Q. Xu, N. Chen, H. Chen, Y. Cheng, F. Zhang, Z. Tan, J. Xing, Q. Chen, J. Zhu, Influence of acceptor–donor codoping on the structure, electrical properties, and hardness of CaBi2Nb2O9-based ceramics, J. Alloys Compd. 910 (2022), 164853, https://doi.org/10.1016/j.jallcom.2022.164853.

[43] X. Du, I.W. Chen, Ferroelectric thin films of bismuth-containing layered perovskites: Part I, $Bi_4Ti_3O_{12}$, J. Am. Ceram. Soc. 81 (1998) 3253–3259, https://doi.org/10.1111/j.1151-2916.1998.tb02764.x.

[44] G. Liu, D. Wang, C. Wu, J. Wu, Q. Chen, A realization of excellent piezoelectricity and good thermal stability in $CaBi_2Nb_2O_9$: Pseudo phase boundary, J. Am. Ceram. Soc. 102 (2019) 1794–1804, https://doi.org/10.1111/jace.16033.

[45] W. Zou, J. Wang, Z. Chen, N. Shi, Z. Li, Z. Cui, X. Li, X. Yin, W. Yan, H. Huang, R. Peng, Z. Fu, Y. Lu, Anisotropic electrical and magnetic properties in grain-oriented $Bi_4Ti_3O_{12}$-$La_{0.5}Sr_{0.5}MnO_3$, J. Mater. Chem. C 6 (2018) 11272–11279, https://doi.org/10.1039/C8TC03809K.

[46] N.A. Zhuk, V.P. Lutoev, B.A. Makeev, S.V. Nekipelov, A.V. Koroleva, A.V. Fedorova, M.V. Yermolina, D.S. Beznosikov, L.O. Karlova, Magnetic susceptibility, EPR, NEXAFS and XPS spectra of Fe-doped $CaBi_2Nb_2O_9$, J. Mater. Res. Technol. 9 (2020) 4173–4182, https://doi.org/10.1016/j.jmrt.2020.02.044.

[47] C.L. Diao, J.B. Xu, H.W. Zheng, L. Fang, Y.Z. Gu, W.F. Zhang, Dielectric and piezoelectric properties of cerium modified $BaBi_4Ti_4O_{15}$ ceramics, Ceram. Int. 39 (2013) 6991–6995, https://doi.org/10.1016/j.ceramint.2013.02.036.

[48] J.D.H. Donnay, D. Harker, A new law of crystal morphology extending the law of bravais, Am. Mineral. 22 (1937) 446–467. https://pubs.geoscienceworld.org/msa/ammin/article-abstract/22/5/446/537845/A-new-law-of-crystal-morphology-extending-the-Law.

[49] D. Aquilano, Basic mechanisms of crystal growth, in: H. Arend, J. Hulliger (Eds.), Crystal Growth in Science and Technology, Springer US, New York, 1989, pp. 49–67, https://doi.org/10.1007/978-1-4613-0549-1_3.

[50] X. Tian, S. Qu, H. Ma, Z. Pei, B. Wang, Effect of grain size on dielectric and piezoelectric properties of bismuth layer structure $CaBi_2Nb_2O_9$ ceramics, J. Mater. Sci. Mater. Electron. 27 (2016) 13309–13313, https://doi.org/10.1007/s10854-016-5480-7.

[51] R. Ahluwalia, T. Lookman, A. Saxena, W. Cao, R. Ahluwalia, T. Lookman, A. Saxena, W. Cao, Domain-size dependence of piezoelectric properties of ferroelectrics, Phys. Rev. B 72 (2005), 014112, https://doi.org/10.1103/PhysRevB.72.014112.

[52] S. Xie, J. Shi, Q. Xu, Q. Wang, J. Zhu, Y. Sato, Q. Chen, In-depth understanding of {110}-type domain walls in bismuth titanate ceramics, Scripta Mater. 217 (2022), 114793, https://doi.org/10.1016/j.scriptamat.2022.114793.

[53] F. Wang, X. Li, Q. Xu, H. Chen, J. Xi, F. Zhang, Z. Tan, Y. Cheng, S. Wang, J. Xing, Q. Chen, J. Zhu, Simultaneous enhancement of electrical and mechanical properties in $CaBi_2Nb_2O_9$-based ceramics, J. Eur. Ceram. Soc. 42 (2022) 4196–4211, https://doi.org/10.1016/j.jeurceramsoc.2022.03.044.

[54] J.Y. Li, R.C. Rogan, E. Üstündag, K. Bhattacharya, Domain switching in polycrystalline ferroelectric ceramics, Nat. Mater. 4 (2005) 776–781, https://doi.org/10.1038/nmat1485.

[55] C.A. Randall, N. Kim, J.P. Kucera, W. Cao, T.R. Shrout, Intrinsic and extrinsic size effects in fine-grained morphotropic-phase-boundary lead zirconate titanate ceramics, J. Am. Ceram. Soc. 81 (1998) 677–688, https://doi.org/10.1111/j.1151-2916.1998.tb02389.x.

[56] X. Wang, J. Wu, D. Xiao, X. Cheng, T. Zheng, X. Lou, B. Zhang, J. Zhu, New potassium-sodium niobate ceramics with a giant d_{33}, ACS Appl. Mater. Interfaces 6 (2014) 6177–6180, https://doi.org/10.1021/am500819v.

[57] T.R. Shrout, S.J. Zhang, Lead-free piezoelectric ceramics: Alternatives for PZT? J. Electroceram. 19 (2007) 113–126, https://doi.org/10.1007/s10832-007-9047-0.

[58] D. Damjanovic, Contributions to the piezoelectric effect in ferroelectric single crystals and ceramics, J. Am. Ceram. Soc. 88 (2005) 2663–2676, https://doi.org/10.1111/j.1551-2916.2005.00671.x.

[59] M.J. Haun, E. Furman, S.J. Jang, L.E. Cross, Thermodynamic theory of the lead zirconate-titanate solid solution system, part i: Phenomenology, Ferroelectrics 99 (1989) 13–25, https://doi.org/10.1080/00150198908221436.

[60] M. Zhang, Z. Chen, Y. Yue, T. Chen, Z. Yan, Q. Jiang, B. Yang, M. Eriksson, J. Tang, D. Zhang, Z. Shen, I. Abrahams, H. Yan, Terahertz reading of ferroelectric domain wall dielectric switching, ACS Appl. Mater. Interfaces 13 (2021) 12622–12628, https://doi.org/10.1021/acsami.1c00523.

[61] D.Y. Suárez, I.M. Reaney, W.E. Lee, Relation between tolerance factor and T_c in Aurivillius compounds, J. Mater. Res. 16 (2001) 3139–3149, https://doi.org/10.1557/JMR.2001.0433.

[62] C. Long, Q. Chang, Y. Wu, W. He, Y. Li, H. Fan, New layer-structured ferroelectric polycrystalline materials, $Na_{0.5}Nd_xBi_{4.5-x}Ti_4O_{15}$: crystal structures, electrical properties and conduction behaviors, J. Mater. Chem. C 3 (2015) 8852–8864, https://doi.org/10.1039/C5TC01237F.

[63] C. Long, Q. Chang, H. Fan, Differences in nature of electrical conductions among $Bi_4Ti_3O_{12}$-based ferroelectric polycrystalline ceramics, Sci. Rep. 7 (2017) 4193, https://doi.org/10.1038/s41598-017-03266-y.

[64] S.H. Yoon, H. Kim, Effect of donor (Nb) concentration on the bulk electrical resistivity of Nb-doped barium titanate, J. Appl. Phys. 92 (2002) 1039–1047, https://doi.org/10.1063/1.1486049.

[65] X. Wu, C. Wu, D. Yang, J. Yin, J. Wu, Strain regulation via composition and valence dependent substitution in BNT-based solid solutions, Adv. Powder Mater. 2 (2023), 100079, https://doi.org/10.1016/j.apmate.2022.100079.

[66] S. Liu, P. Gao, H. Zou, B. Qin, J. He, L. Deng, Large magnetoelectric effect in $BaFe_{12}O_{19}$-($Ba_{0.85}Ca_{0.15}$)($Zr_{0.1}Ti_{0.9}$)O_3 particulate composite, Adv. Powder Mater. 1 (2022), 100022, https://doi.org/10.1016/j.apmate.2021.12.001.

[67] H. Chen, X. Guo, Z. Cui, J. Zhai, Donor and acceptor doping effects on the electrical conductivity of $CaBi_2Nb_2O_9$ ceramics, Phys. Status Solidi A 210 (2013) 1121–1127, https://doi.org/10.1002/pssa.201228804.

[68] H. Zhang, H. Yan, M.J. Reece, The effect of Nd substitution on the electrical properties of Bi_3NbTiO_9 Aurivillius phase ceramics, J. Appl. Phys. 106 (2009), 044106, https://doi.org/10.1063/1.3205006.

[69] Z. Zhou, X. Dong, H. Yan, H. Chen, C. Mao, Doping effects on the electrical conductivity of bismuth layered Bi_3TiNbO_9-based ceramics, J. Appl. Phys. 100 (2006), 044112, https://doi.org/10.1063/1.2245210.

[70] Z.Y. Shen, H. Sun, Y. Tang, Y. Li, S. Zhang, Enhanced piezoelectric properties of Nb and Mn co-doped $CaBi_4Ti_4O_{15}$ high temperature piezoceramics, Mater. Res. Bull. 63 (2015) 129–133, https://doi.org/10.1016/j.materresbull.2014.11.055.

[71] Z.Y. Shen, H. Sun, Y. Tang, Y. Li, S. Zhang, Enhanced piezoelectric properties of Nb and Mn co-doped $CaBi_4Ti_4O_{15}$ high temperature piezoceramics, Mater. Res. Bull. 63 (2015) 129–133, https://doi.org/10.1016/j.materresbull.2014.11.055.

[72] J. Chen, J. Yuan, S. Bao, Y. Wu, G. Liu, Q. Chen, D. Xiao, J. Zhu, Effects of (Li, Ce, Y) co-substitution on the properties of $CaBi_2Nb_2O_9$ high temperature piezoceramics, Ceram. Int. 43 (2017) 5002–5006, https://doi.org/10.1016/j.ceramint.2017.01.007.

[73] G. Liu, J. Yuan, R. Nie, L. Jiang, Z. Tan, J. Zhu, Q. Chen, Electrical properties and thermal stability of Ce-modified $Ca_{0.80}(Li_{0.5}Bi_{0.5})_{0.20}Bi_2Nb_2O_9$ ceramics, J. Alloys Compd. 697 (2017) 380–387, https://doi.org/10.1016/j.jallcom.2016.11.403.

[74] H.S. Ma, M.K. Lee, B.H. Kim, K.H. Park, J.J. Park, S.H. Lee, Y.G. Jeong, K.I. Park, C.K. Jeong, G.J. Lee, Role of oxygen vacancy defects in piezoelectric thermal stability characteristics of Mn-doped (K,Na,Li)NbO_3 piezoceramics, Ceram. Int. 47 (2021) 27803–27815, https://doi.org/10.1016/j.ceramint.2021.06.207.

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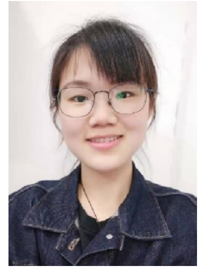

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